

Line wrap

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1 <!DOCTYPE html>
2 <html lang="en">
3
4 <head>
5   <meta charset="UTF-8">
6   <!-- 📱 MANDATORY FOR RESPONSIVE: Viewport meta tag tells mobile browsers not to zoom out or pretend to be a desktop display -->
7   <meta name="viewport" content="width=device-width, initial-scale=1.0">
8   <title>MCC ADD 103 | Sample 07B: Responsive Grid Layout</title>
9   <!-- 🌐 MCC Fonts & External Stylesheet -->
10  <link rel="stylesheet" href="style.css">
11 </head>
12
13 <body>
14
15   <!-- 🏠 MCC BRAND HEADER -->
16   <header class="mcc-header">
17     <span class="mcc-badge">McHenry County College – ADD 103</span>
18     <h1 class="page-title">📱 Responsive Grid Layout</h1>
19     <p class="page-subtitle">Resize your browser window to see cards adapt from 1 column on mobile to 2 columns on tablet and 4 columns on desktop!</p>
20   </header>
21
22   <main class="container">
23
24     <!-- 📱 MOBILE-FIRST GRID: This container starts as a single column stack on mobile screens, then expands at 600px and 900px -->
25     <div class="card-container">
26
27       <!-- 📱 Item 1: Displays in 1 column on mobile, 2 columns on tablet, 4 columns on desktop -->
28       <div class="card">
29         
30         <div class="content">
31           <h2>Mobile First</h2>
32           <p>This layout starts as a single column stack on mobile devices.</p>
33         </div>
34       </div>
35
36       <!-- 📱 Item 2: Moves next to Item 1 at the 600px tablet breakpoint -->
37       <div class="card">
38         
39         <div class="content">
40           <h2>Tablet View</h2>
41           <p>At 600px, it splits into a 2-column layout.</p>
42         </div>
43       </div>
44
45       <!-- 📱 Item 3: Expands into a 4-column row at the 900px desktop breakpoint -->
46       <div class="card">
47         
48         <div class="content">
49           <h2>Desktop View</h2>
50           <p>At 900px, it expands to a 4-column row.</p>
51         </div>
52       </div>
53
54       <!-- 💧 Item 4: Fluid design lets all cards resize seamlessly with fraction units (1fr) -->
55       <div class="card">
56         
57         <div class="content">
58           <h2>Fluid Design</h2>
59           <p>Fraction units (1fr) adapt smoothly to available screen width.</p>
60         </div>
61       </div>
62
63     </div>
64
65     <!-- =====>
66     📖 STUDENT GUIDE: Media Queries & Responsive Design Explained
67     <!-- ===== -->
68     <section class="explanation-section">
69       <h2>
70         💡 What Are Media Queries & How Do They Work?
71       </h2>
72
73       <h3>1. The Simple Analogy</h3>
74       <p>
75         Imagine dressing for the weather. If it's hot outside, you wear a t-shirt. If it starts raining, you put on a raincoat.
76         <strong>Media queries</strong> work the same way for websites! They ask the browser: <em>"How wide is this screen right now?"</em>
77         Based on the answer, the browser applies different CSS styling rules.
78       </p>
79
80       <h3>2. Why "Mobile-First"?</h3>
81       <p>
82         In modern web development, we usually write CSS <strong>mobile-first</strong>:
83       </p>
84       <ul>

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85         <li><strong>Base CSS (Default):</strong> Write styles for mobile screens first (e.g., a simple 1-column layout).</li>
86         <li><strong>Media Queries:</strong> As the screen gets wider (tablets, laptops, desktops), media queries kick in to expand the layout i
87     </ul>
88
89     <h3>3. Anatomy of a Media Query</h3>
90     <p>Here is what the code looks like in your CSS file:</p>
91     <pre class="code-block"><code>/* 1. Base styles (apply to mobile phones by default) */
92     .card-container {
93         display: grid;
94         grid-template-columns: 1fr; /* 1 Column */
95     }
96
97     /* 2. Tablet view: triggers at 600px width or wider */
98     @media (min-width: 600px) {
99         .card-container {
100             grid-template-columns: 1fr 1fr; /* 2 Columns */
101         }
102     }
103
104     /* 3. Desktop view: triggers at 900px width or wider */
105     @media (min-width: 900px) {
106         .card-container {
107             grid-template-columns: repeat(4, 1fr); /* 4 Columns */
108         }
109     }</code></pre>
110
111     <h3>4. The Secret Ingredient: Viewport Meta Tag</h3>
112     <p>
113         Media queries won't work on mobile devices without this required line inside your HTML <code>&lt;head&gt;</code>:
114     </p>
115     <pre class="code-block-highlight"><code>&lt;meta name="viewport" content="width=device-width, initial-scale=1.0"&gt;</code></pre>
116     <p>
117         Without it, phone browsers pretend to be desktop screens and zoom out, making your site tiny and unreadable!
118     </p>
119
120     <h3>5. How to Test It Yourself</h3>
121     <ul>
122         <li><strong>Resize your window:</strong> Click and drag the corner of your browser window to make it wider or narrower. Notice how the
123         <li><strong>Use Developer Tools:</strong> Press <code>F12</code> or right-click &rarr; <code>Inspect</code>, then click the <strong>Device To
124     </ul>
125 </section>
126
127 </main>
128
129 <!-- 🏠 MCC FOOTER -->
130 <footer class="mcc-footer">
131     <p>Created by Dr. Meri Kasprak in AntiGravity with collaboration from GitHub Copilot (Gemini 3.6 Flash).</p>
132     <p>McHenry County College • ADD 103 Web Development</p>
133     <p><a href="./../index.html">🏠 Course Home Page</a></p>
134 </footer>
135
136 </body>
137
138 </html>
139
140 </html>

```